EECS 498 Task 8

How I Located The Microphone

I located the microphone by researching my specific M2 MacBook Air. The microphone and speakers are near the screen hinge, as most of the MacBook Airs are designed. Thus, there was difficulty with hand placement and testing. However, I saw subtle changes in the spectrogram.

Visualization Results for Task 8

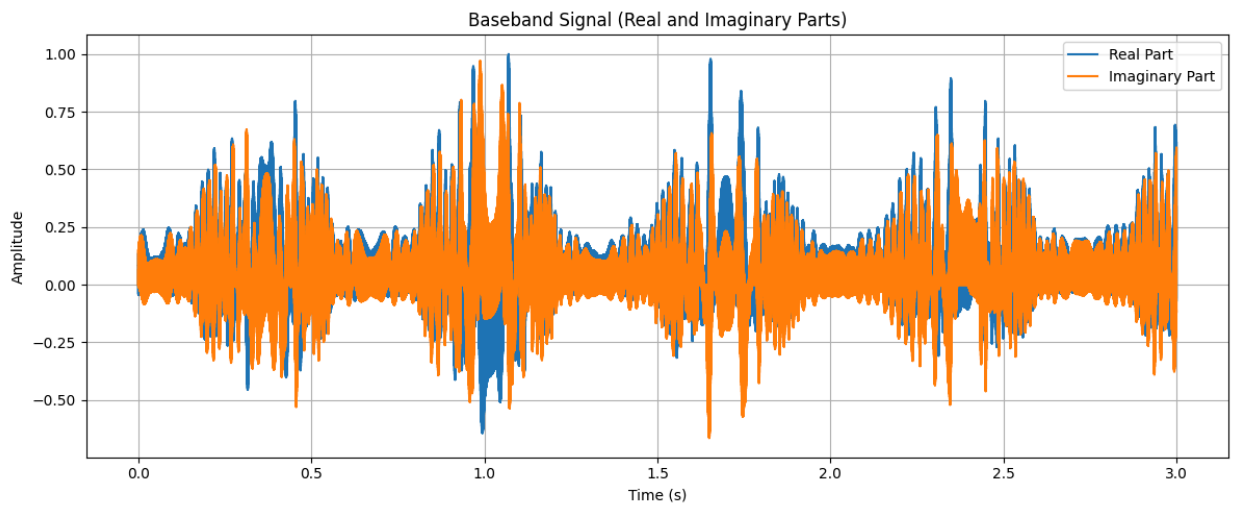

Figure 1. This is the non-filtered, down-converted baseband signal of my data.

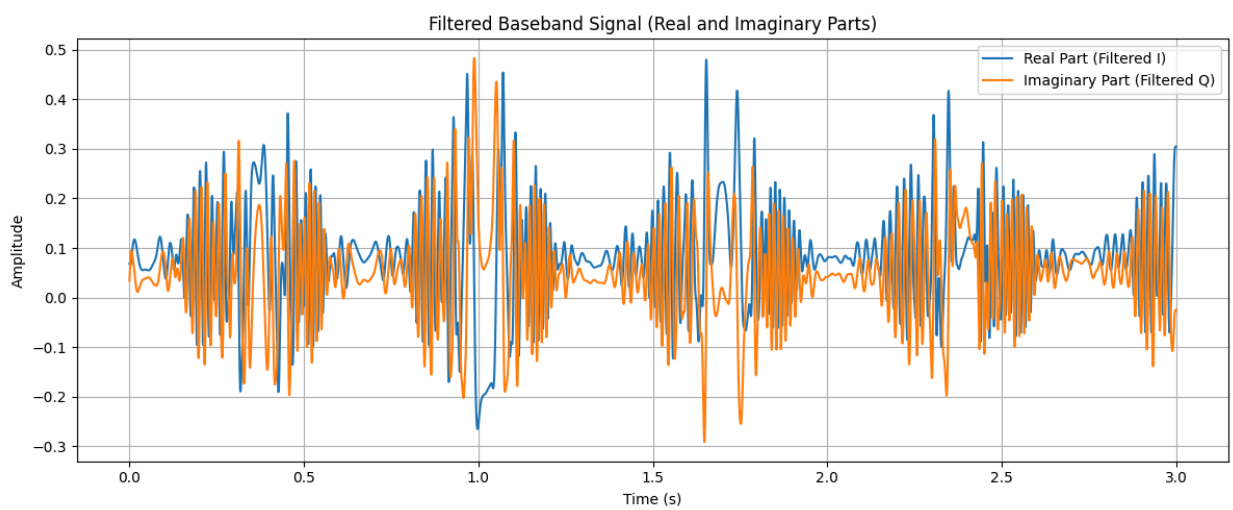

Figure 2. The signal is passed through a 5th-order low-pass filter.

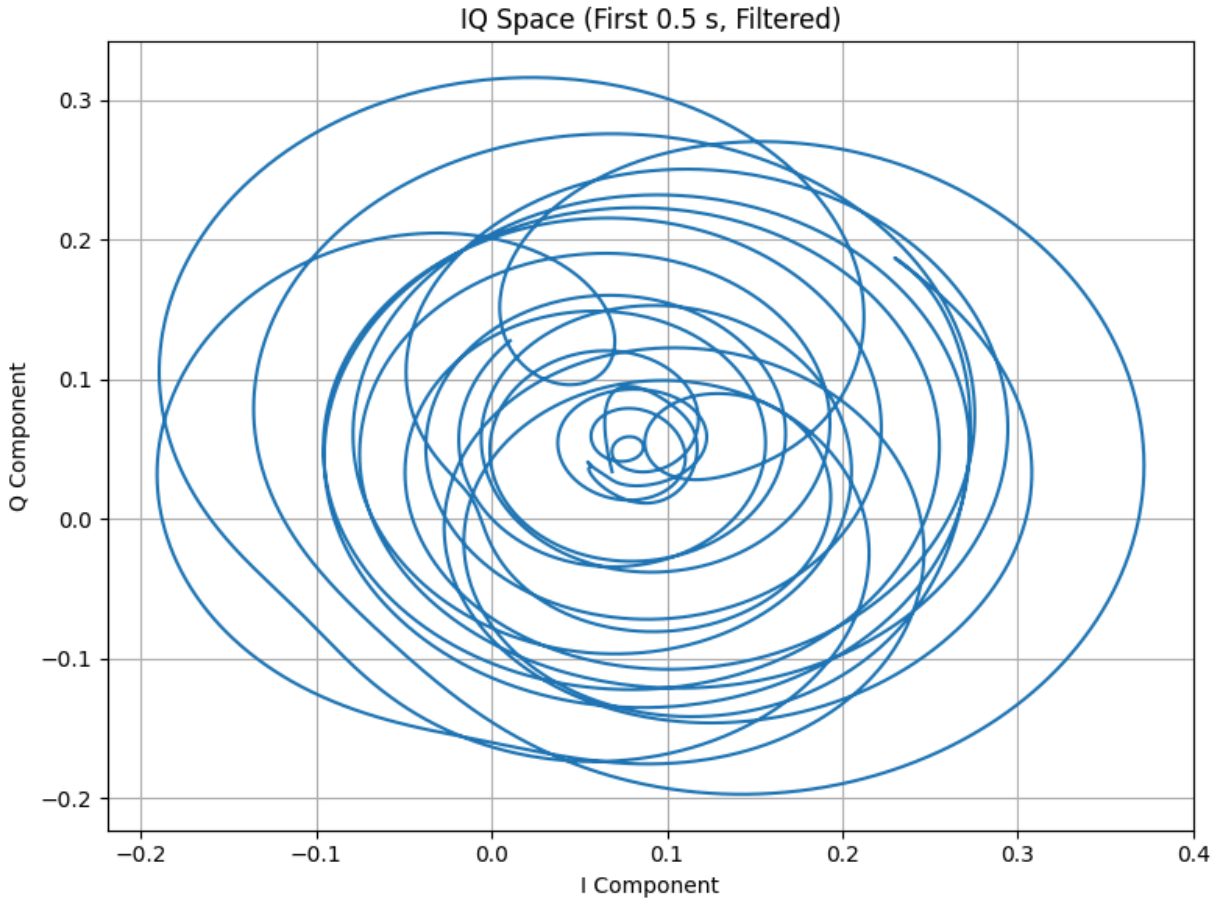

Figure 3. The 2D phase change in the I/Q domain.

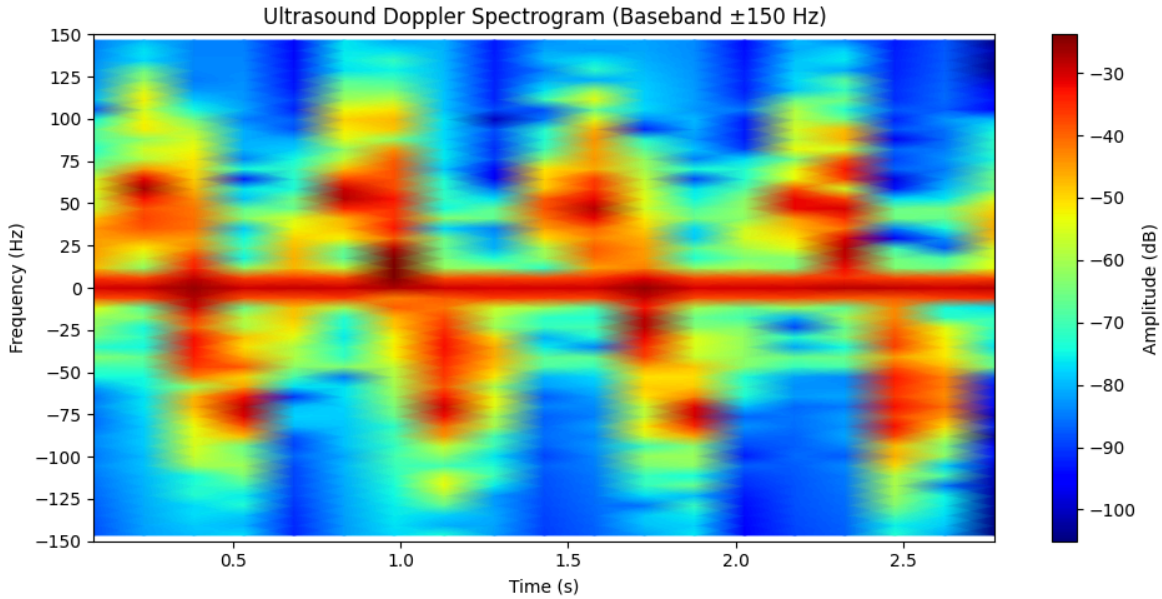

Figure 4. The Doppler Spectrogram of the downsampled filtered signal